

1. What is a Database? Define it in your own words and explain why databases are essential in the modern world. Provide two real-world examples of where databases are used- **Database is an organized collection of structure information or data. Banking systems to store account information and transaction history of the customers. Insurance companies uses to data to assess clients risks and determine the appropriate premiums for the policies**

2. What is a Data Warehouse? **Data Warehouse is a large central location where data is managed and stored for analytical processing**

- How does it differ from a regular database?

	Data warehouse	Database
Purpose	Analysis	Reporting and security
Database	OLAP	OLTP

- Why would a company use a data warehouse?- **To improve data quality and enhanced decision support**

3. What is a Data Lake? **It's a centralized storage system that stored structured, semi structured and unstructured data in its raw format for flexible analysis.**

- Explain the concept and describe how it differs from a data warehouse.

Feature	Data Lake	Data warehouse
Data Type	Structured, semi-structured, and unstructured data	Structured data only
Governance	Required strong external governance	Built-in structure and control
Primary use	Big data, AI, ML and real time analysis	Business Intelliegence

- When would a business prefer a data lake over a data warehouse? **when it needs flexible, low-cost storage for raw, multi-format data to support machine learning, advanced analytics, or future, undefined use cases.**

4. What is a Data Mart? **It's a data storage system that contains information specific to an organization business unit, it provides summarized data that key stakeholders can use to quickly make informed decisions.**

- How does it relate to a data warehouse and who typically uses it: **It's a smaller specific subset of data warehouse that serves the needs of a specific department or business unit like Marketing/ Sales/ Finance.**

5. Explain the difference between Structured, Semi-Structured, and Unstructured Data. Provide one example of each.

Type of Data	Define	Example
Structured Data	Data that is organized that follows a predefined schema	Financials transactions (Bank deposits), Point of sales (Sales records)
Semi-Structured Data	Data that does not reside in a relational database but that has some organizational properties that make it easier to analyze.	XML or JSON
Unstructured Data	Unstructured data lacks a predefined format or schema, making it difficult to store in traditional relational databases.	Social media, Videos and Images

6. What is Data Modeling? **Is the practice of designing how data is structured and related so that it can be stored, understood and analyzed consistently.**

- Why is it important when designing a system that works with data?
- Describe at least two types of data models- **Logical and Conceptual data model**

7. What is Data Mining? **Systematic process of organizing large datasets to identify patterns, correlation and trends.**

- How is it used in business? **They use it to define rules that can be used to design a new system or adapt an existing one and to align business and technology goals.**

- Provide two real-world examples of data mining in action- **Fraud Detection and customer segmentation.**

8. What is a Database Management System (DBMS)? **It's a collection of multiple software services that work together to store, maintain, structure and deliver the data as part of a product.**

- Name and briefly describe four popular DBMS platforms (e.g., MySQL, PostgreSQL, Oracle, MongoDB) and explain a suitable use case for each

DBMS platforms	Description	Case
MySQL	Is an open-source relational database management system that uses SQL for managing and manipulating data	Ideal for web applications, e-commerce, and blogging platforms
PostgreSQL	Is an open-source relational database management system (RDBMS) known for its robustness, extensibility and compliance with SQL standards.	Suitable for complex applications needing reliability and data integrity
MongoDB	Is a NoSQL database management system designed to handle unstructured and semi-structured data.	Best for big data, content management, and real-time analytics
Oracle	Is a relational database management system created and run by the Oracle Corporation.	Large-scale enterprise applications, Financial institutions

9. What is ETL (Extract, Transform, Load)? **Is a process that extracts the data from different source systems, then transforms the data loads the data into the data warehouse system.**

- Explain each step and describe why this process is important in data analytics
 - **STEP 1: Extraction-** Data is extracted from the source system into the staging area
 - **STEP 2: Transformation-** data in the staging area is transformed through the [data processing](#) phase to make it suitable for use for analytics
 - **STEP3: Loading-** The load phase moves the transformed data into a permanent target system
- It's important in data analytics -Consolidate data from multiple, different sources into a single system. And it improves data quality by cleaning and standardizing information for accurate analytics and reporting.

10. What are common data storage formats used in analytics (e.g., CSV, JSON, Parquet, Avro)? Briefly describe each and state when you would use one over another

- **CSV (Comma-Separated Values)-** Is the simplest form of tabular data and it used for basic data exchange between different applications, simple datasets.
- **JSON (JavaScript Object Notation)-** is a lightweight, humana readable data interchange format. It is used for semi-structured data, flexible schemas and applications that are needed to easily represent hierarchical data.
- **Parquet (Columnar storage format)-** is a columnar storage file optimised for use with big data processing frameworks like Apache Spark and Hadoop. It's a go-to format for big analytics, data lakes and when performance for analytical queries id critical.
- **Avro (Row-based storage format)-** is a row-based remote procedure call and data serialization framework developed within Apache Hadoop. Its used for data serialization in big data environments, especially with Apache Kagka for streaming data

11. What is Artificial Intelligence (AI)? Define it and explain the difference between Narrow AI (Weak AI) and General AI (Strong AI). Provide one example of each.

- **Artificial intelligence is a technology that enables machines and computers to perform tasks that typically require human intelligence.**
- **Narrow AI-** It's an AI that has ability to perform very specialized tasks. Example- Self driving cars and smart assistant like SIRI and Alexa

- **General AI- It's an AI that can perform actions in the same as human beings can. Example- Ability to reason and solve a puzzle**

12. What is Machine Learning (ML)? How does it differ from traditional programming?

Describe the three main types of Machine Learning: Supervised, Unsupervised, and Reinforcement Learning.

- **Machine Learning- is a subset of artificial intelligence that enables computers to learn from data and make predictions without explicit programming**
- **Machine Learning differs from Traditional programming because it learns from data rather than relying solely on manual coded rules**
- **Supervised- the model is trained on a labeled dataset.**
- **Unsupervised- Discovering patterns in unlabeled data.**
- **Reinforcement- Learning through interactions with an environment.**

13. What is Deep Learning? How does it relate to Machine Learning and Artificial Intelligence? What are Neural Networks and how do they work?

- **Deep learning- Subset of machine learning in which multilayered neural networks learn from vast amounts of data**
- **AI manages more comprehensive issues of automating a system while Machine learning manages to influence users' machines to gain from the external environment**
- **Neural Networks- is an algorithm designed to recognize patterns. It's inspired by how the human brain processes information.**
- **Neural Networks feed in data, weight processing, activate functions, output generation and learn from mistakes**

14. What is Natural Language Processing (NLP)? Describe what it is and provide three real-world applications of NLP that you use in everyday life

- **Natural Language Processing (NLP)- it's a subset of artificial intelligence, computer science and linguistics focused on making human communication.**
- **Three real-world applications of NLP that you use in everyday life- Voice-activated assistants, Machine translation, Chatsbots**

15. What is a Large Language Model (LLM)? - It's an AI system trained on massive amounts of text data to understand and generate human-like language

- How is it different from a standard Machine Learning model? **ML is broad field used for classification, regression, clustering and forecasting across many domains, while LLM is a specialized branch focused on understanding and generating human language.**
- Name two well-known LLMs and explain what they are used for
 - **GPT-4- Is a state-of-the-art LLM developed by OpenAI, it's a model capable of processing both text and image inputs**
 - **Claude- designed with a focus on safety, clarity and alignment with human values**

16. What is Generative AI? It's an AI Systems that can create new contents like text, images, Music by learning patterns from existing data.

- How does it work at a high level? – **Works by using machine learning model. It works in Training, Tuning and generation, evaluation and retuning**
- Name three tools that use Generative AI and describe what each one does.
 - **ChatGPT: Creative writing assistance and chatbots and can be applied in Customer support and content creation**
 - **Bard (Gemini): Creative writing, coding, solving math problems and can be used by writers and educators for content creation**
 - **GitHub Copilot: Code completion and suggestions and it assists developers in writing code efficiently in IDEs like Visual Studio Code**

17. What is a Training Dataset? Collection of examples that show an AI and ML models how to make predictions, recognize patterns or generate content.

- Why is the quality of training data so important when building an AI model?
 - **Data quality refers to how Accurate, Consistent, Complete and relevant data is across the entire lifecycle**
- What could go wrong if the training data is biased or incorrect?
 - **It may produce inaccurate predictions or decisions because they learn patterns that do not reflect real world conditions**

18. What is Computer Vision? Is a field of AI that enables computers to see, identify, and understand visual information from images and videos, similar to how humans see and process this information

- How do machines 'see' and interpret images?
 - **Visual data is first captured through cameras or sensors, then cleaned and standardized so it can be analyzed consistently. From there, machine learning models extract features and generate predictions based on patterns learned during training.**
- Provide two real-world use cases of computer vision
 - **Face recognition for unlocking phones**
 - **Self-driving cars**

19. What is an API (Application Programming Interface)? It's a set of rules or protocols that enables software applications to communicate with each other to exchange data, features and functionality.

- Explain what it is using a simple real-world analogy.
 - **Imagine you're at a restaurant: The menu is the API — it shows what you can order. You (the client) tell the waiter (API) your order. The waiter brings your food from the kitchen (server).**
- Why are APIs important in modern software and data systems?
 - **They are essential in modern software because they act as bridges that allow different applications, services, and devices to communicate seamlessly, regardless of their underlying technologies.**
 - **Interoperability APIs enable different systems to work together without needing to be built in the same programming language or environment**
 - **Efficiency and Reusability APIs allow developers to reuse existing functionalities instead of reinventing them, saving significant time and resources**

20. What is an API Key? Is a unique series of letters and numbers that's a special ID or secret password for an application or user when making a call to an API

- What is its purpose?
 - **Control Access:** API keys serve as gatekeepers. They can only be accessed by valid keys, thus allowing only legitimate individuals or apps to access the data.
 - **Security Layer:** An API key is a simple security mechanism like a lock on a door. It ensures that sensitive information is safeguarded by preventing everyone from gaining entry.
 - **Usage Tracking:** API keys allow service providers to track the usage of their API.
- How should it be stored and protected, and what could happen if it is exposed publicly?

Storage Method	Security Level
Source Code	Very Poor
Configuration File	Poor
Environment Variables	Good
Secret Manager	Excellent
Hardware Security Module (HSM)	Highest

- If its exposed publicly it can cause Data breaches, financial loss, Service disruption and regulatory penalties

21. What is a REST API? Is an application programming interface (API) that conforms to the design principles of the representational state transfer (REST) architectural style, a style used to connect distributed hypermedia systems.

- What does REST stand for and what are the main HTTP methods (GET, POST, PUT, DELETE) used in REST APIs?
 - REST stand for **REpresentational State Transfer**

- Briefly describe each method:
 - **GET:** The get method is used to Read data on the server.
 - **POST:** The post method is used to Create data.
 - **PATCH/PUT:** The patch method is used to Update data.
 - **DELETE:** The delete method is used to Delete data.

22. What is JSON (JavaScript Object Notation)? It's a lightweight data interchange format that is easy for humans to read and write and easy for machines to analyse and generate.

- Why is it commonly used when working with APIs and data exchange?
 - **It is language-independent**
 - **It is easy to read while debugging**
 - **It works naturally with the web**
 - **It is usually less wordy than XML**
- Write a simple example of a JSON object.

```
{  
  "name": "Dineo Mogami",  
  "age": 30,  
  "email": "dineomogami@gmail.com",  
  "isActive": true,  
  "tags": ["developer", "javascript", "json"],  
  "address": {  
    "street": "15th road",  
    "city": "Johannesburg",  
    "country": "SA"  
  }  
}
```

23. What is Prompt Engineering? Is the process where you guide Generative AI solutions to generate desired output.

- Why is it an important skill when working with AI tools like ChatGPT or Claude?
 - Improved user experience; users avoid trial and error and still receive accurate and relevant responses from AI Tools.
 - Increased flexibility, higher levels of abstraction improve AI models and allow organizations to create more flexible tools at scale.

- **Greater developer control- Effective prompts provide intent and establish context to the large language models.**

- Provide two examples of a poorly written prompt and an improved version of each.

Bad Prompt	Good Prompt
Explain gravity.	Explain gravity to a 12-year-old using the example of a ball falling from a table.
Solve a math problem.	Solve $2x + 5 = 15$ step by step and explain each step clearly.

24. What is a MAP function? **It's function in python that applies a function to every element of one or more iterables and returns a map object containing the transformed results.**

- Explain what it does in programming and data processing.
 - **Map Phase: Data is split into smaller chunks and processed in parallel across multiple nodes, transforming the input data into intermediate key value pairs**
 - **Reduce phase: This model is designed to handle large datasets efficiently, leveraging parallel processing and fault tolerance to ensure data integrity and speed.**
- Provide a simple example in either Python or JavaScript.

```
def double(val):  
    return val * 2
```

```
a = [1, 2, 3, 4]
res = list(map(double, a))
print(res)
```

- Output [2, 4, 6, 8]

25. What is Cloud Computing? It's process where technology services like storage, databases, computing power, and software are delivered over the internet.

- Name the three main cloud service models (IaaS, PaaS, SaaS) and briefly explain each. Give one example of a product for each model.
 - **IaaS (Infrastructure as a Service)- It's a renting computer online, you can practice coding or run website without buying any hardware. Example: Rent a powerful server from AWS**
 - **PaaS (Platform as a service)- It's a model that give you the tools to build and launch projects. Example: Building of Mobile App**
 - **SaaS (Software as a Service)- It's a software that you can directly use on the internet. Example: Write documents using Google Docs**

26. What is a Version Control System? It's a system that records changes to a file or set of files over time so that you can recall specific versions later.

- What is Git and why is it widely used by data analysts and developers?
 - **Git is a version control software that tracks changes to files within a folder that you assign a Git to track.**
 - **It is essential for Data analysts and Developers for tracking changes to SQL Queries, Python scripts, R scripts and notebooks. It allows multiple developers to work on the same project, view previous versions, and restore changes when needed.**
- Briefly explain what a 'commit' and a 'branch' are.
 - **a'commit'-a snapshot created at a specific moment, with a unique SHA-1 hash, immutable once created**
 - **a 'branch'- line of development that traces back through commits, each commit pointing to its parent**

27. What is a Jupyter Notebook? **is a web-based application popular in data science and machine learning that you can use to record, create, and share computational documents like live code, equations, and visualizations.**

- What programming language is it most commonly used with? **Java, Scala, MATLAB,**
- Why is it popular among Data Analysts and Data Scientists?
 - **Interactive Computing:** By executing code in separate cells, users can test and improve their code without executing the complete script.
 - **Supports Multiple Languages:** Jupyter's kernel system allows it to support languages like R and Julia, even though it is predominantly used for Python.
 - **LaTeX syntax:** The implementation of LaTeX syntax in Jupyter notebooks allows mathematicians and physicists to easily write and document equations.
 - **Data visualizations:** Jupyter Notebook uses Python's data visualization libraries, such as Matplotlib, Seaborn, and Plotly, to easily generate graphs and charts in your notebook.

28. What is Python? Why is it considered one of the most popular programming languages for Data Analytics and AI? **Python is a high-level, interpreted programming language that has gained widespread recognition for its simplicity and readability. Data analysts often work with messy spreadsheets, large datasets, and complex financial models. Python libraries like NumPy, Pandas, Matplotlib, and Seaborn give you simple commands to handle those tasks quickly and accurately. Instead of spending hours writing low-level code, you can focus on interpreting the data and making decisions.**

- Name four Python libraries commonly used in data work and briefly describe each
 - **Pandas**
 - **NumPy**
 - **Matplotlib**
 - **Seaborn**

29. . What is Data Privacy? Why is it important? Data privacy refers to the right of individuals to control how their personal information is collected, stored, shared, and used. It ensures that individuals maintain authority over the data they generate and that organizations process it responsibly. The importance of data privacy extends across multiple domains: personal rights, business ethics, societal well-being, and legal compliance.

- Research the GDPR (General Data Protection Regulation) and explain what it is and which countries it applies to.
 - **European Union (EU) law surrounding data privacy and security, which was adopted by the European Parliament in 2016 and came into effect in May 2018. The legislation marked a major worldwide precedent; it was particularly notable for establishing a comprehensive framework around data protection, expanding the legal definition of personal data, and extending its protection to all data generated by EU citizens and residents regardless of where the information was being processed.**
 - **Other core principles of the GDPR include purpose limitation (data must be collected and processed for a legitimate purpose), data minimization (the scope of the data must be limited to what is required for the organization's purpose), storage limitation (data must be retained only as long as required), data accuracy (inaccurate data must be rectified or erased), data integrity and privacy (the integrity and confidentiality of the data must be protected at all times), and accountability (proof of compliance).**
 - **GDPR applies directly in all 27 EU Countries, 3 EEA(Norway, Iceland and Liechtenstein) and the United Kingdom.**

30. What is Data Ethics? refers to the principles behind how organizations gather, protect, and use data. It's a field of ethics that focuses on the moral obligations that entities have (or should have) when collecting and disseminating information about us

- Provide three examples of unethical use of data or AI, and explain what makes each example problematic
 - **AI in Job Applications: Resume Writing with AI Tools- Risk of overpromising and underdelivering. Misrepresentation can damage your reputation.**

- **AI in Education: Writing an Essay with ChatGPT- The work is not from the student thinking or writing, the essay might have errors or misinterpretations**
- **AI in Software Development: Coding with GitHub Copilot- The code might have unnoticed security flows or logic errors**
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31. What is Data Governance? Define it and explain the key components of a good data governance framework- Is the data management discipline that focuses on the quality, security and availability of an organization's data. Data governance helps ensure data integrity and data security by defining and implementing policies, standards and procedures for data collection, ownership, storage, processing and use. The goal of data governance is to maintain safe, high-quality data that is easily accessible for data discovery, business intelligence initiatives and AI efforts.

A data governance framework details an organization's structures and processes for managing critical data assets. It defines data ownership and responsibilities and specifies how data should be handled to maintain data quality standards, security and compliance.

32. What is Bias in AI? Occurs when a system produces results that favor or discriminate against certain groups of people.

- **Explain what it means for an AI model to be biased- is when artificial intelligence models produce skewed or unfair outcomes**
- **Provide a real-world example of AI bias and describe the potential consequences.**
 - **Facial recognition bias- machine learning algorithms trained on large datasets of facial images. Bias arises primarily when these datasets are non-representative, often over-representing lighter-skinned individuals and under-representing women and people of color**

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33. What are the key differences between a Data Analyst, a Data Scientist, a Data Engineer, and an AI Engineer? Create a simple table comparing their roles, skills, and tools

Aspect	Data Analyst	Data Scientist	Data Engineer	AI Engineer
Roles	Analyzing data to uncover insights and trends	Building predictive models and solving complex problems	Designing and maintaining data infrastructure	Develops, implements, and maintains AI-powered applications. System Integration & Product Engineering
Skills	SQL, Excel, Data Visualization, Basic Statistics	Python/R, Machine Learning, Statistics, Data Mining	QL, Python, ETL Tools, Cloud Platforms, Database Management	Python, R, Java, or C++. Machine Learning & Deep Learning
Tools	Tableau, Power BI, Excel, SQL	Python, R, TensorFlow, Scikit-learn, Jupyter	Apache Spark, Hadoop, Airflow, Snowflake, AWS, Azure	Pandas, NumPy, A TensorFlow, PyTorch, XGBoost, LightGBM pache Spark

34. What is Business Intelligence (BI)? **Refers to the processes and tools organizations use to analyse their business data, turn it into actionable insights, and help everyone make better-informed decisions and achieve KPIs**

- Name two popular BI tools (e.g., Power BI, Tableau) and explain how they help businesses make data-driven decisions
 - Power BI- **It excels at taking disparate data sources—from simple Excel spreadsheets to complex cloud databases—and turning them into coherent, visually immersive and interactive insights.**
 - Tableau- **business intelligence tool that specializes in data visualization and analysis. Its drag-and-drop interface allows**

businesses to create complex visualizations, interactive dashboards and detailed reports without needing to write code.

35. Which concept from this assignment did you find most interesting or surprising?

Explain why: **Data Privacy in an organization is very crucial if is not followed it can lead to severe consequences leading to financial penalties and legal repercussions and reputational damage.**

36. Which concept did you find most difficult to understand? What resources did you use to help you understand it better? **The concept I found most difficult to understand was the assignment because it covered many new data-related terms. At first, it felt overwhelming, but separating the work into sections per module made it easier to follow and understand each concept step by step. Instead of learning everything at once, I could focus on one topic at a time and connect the ideas more clearly. I used online resources, class notes, and examples from the assignment to help me understand the topics better.**

37. How do you see Data and AI affecting your future career or everyday life? Be specific with examples:

- **I see Data and AI having a big impact on my future career and everyday life because they are becoming part of almost every industry. In my future career, data can help me make better decisions by identifying patterns, trends, and risks. For example, in insurance or business, data can be used to understand customer behaviour, assess risk, detect fraud, and improve decision-making. AI can also help by automating repetitive tasks, such as sorting information, analysing large amounts of data, or generating reports faster.**
- **In my everyday life, I already use AI in tools such as Google Maps for directions, online banking apps for fraud alerts, recommendation systems on Netflix or YouTube, and chatbots for customer service. In the future, I think AI will help me work more efficiently, learn new skills faster, and solve problems more accurately. However, I also believe it is important to use Data and AI responsibly by protecting personal information, avoiding bias, and checking that AI-generated answers are accurate.**

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